Structural Elicitation in a Frozen POSIX Agent: Environmental Affordance Fields for Artificial Agency

Hao Ke[1] Jingyun Wu[1]

[1]Independent Researcher
i@kehao.me joeywu021@gmail.com

Abstract

Artificial life asks how agency arises from the coupling of an agent, its substrate, and a structured environment, rather than from a designer-supplied goal. We introduce a minimal POSIX assay for this coupling: a frozen, instruction-tuned LLM activated as a long-running process with a shell and full autonomy, given no task and no reward. Using matched-pair experiments scored only on un-fakeable filesystem and exec events, we find that the static structure of the workspace directs what the agent does. Across five motifs spanning discrepancy and affordance structure—a broken file repaired, a missing element created, a dangling reference closed, a version conflict reconciled, an executable run—the structure arm produces the matched act in 96 of 102 runs while an otherwise-identical control never does (0/85), for three independent model families (gpt-5.4-codex, deepseek-v4-pro, glm-5.2). The pull replicates on non-code prose across families and persists when the substrate blocks realization. Specific tasking overrides it completely (9/10 to 0/10 on gpt, with GLM repeating 5/5 to 0/5), while a small pilot suggests weak underspecified prompting still leaves the pull intact. The environment does not create a goal ex nihilo; it selects among latent action priors in an activated policy, producing behavior that is purpose-like without being purposive. We report a phenomenon and a cheap, un-fakeable assay for environment-coupled behavior; mechanism is left to future work.

Submission type: **Late Breaking Abstracts**
Data/Code available at: https://github.com/kehao95/quine

Problem

Artificial life studies how agency arises from the *coupling* of an agent, its substrate, and a structured environment, rather than from a designer-supplied goal. For biological agents the environment is a behavior source in its own right — affordances that direct an already-active organism without supplying its goals (Gibson, 1979; Rietveld and Kiverstein, 2014; Barker, 1968). We bring this question to a contemporary artifact: a frozen, instruction-tuned large language model coupled to a real operating-system substrate. The question is deliberately minimal: with no task supplied and no reward given, does the static structure of the environment do anything beyond serve as context — can it direct the agent's functional behavior on its own?

This cuts against how language-model agents are usually studied. The dominant paradigm is *instruction-capture*: a prompt supplies the goal and the environment is context, benchmark, or harness (Park et al., 2023; Wang et al., 2023; Rahwan et al., 2019). Remove the instruction and the question becomes what, if anything, governs a frozen policy in the underspecified regime. Two recent studies border this regime: Szeider (Szeider, 2025) reports spontaneous metacognitive patterns in unsupervised LLM agents but reads behavior off narration rather than a functional substrate, with no matched controls; Ye et al. (Ye et al., 2026) study autonomous exploration that remains goal-directed. Neither asks whether *static environmental structure alone* directs unfakeable behavior. We use Quine (Ke, 2026), a medium in which an LLM runs as a real POSIX process, so that behaviour is read off *un-fakeable* filesystem and exec events rather than narration.

Setup

A single frozen-weight LLM is activated as a long-running POSIX process with a shell and one standing prime: *"You are a running process in a workspace, with sh available. You have full autonomy and agency,"* and an opening user turn, *"No instructions for you; you may inspect the workspace first then act freely."* No task is supplied; no reward is given during the episode; weights are fixed. This is a *no-task* condition, not a no-prompt condition — the prime supplies activation, but no goal. The operating system is the substrate: file permissions, read-only mounts, and kernel-level write refusal are not simulated but real, giving the agent a genuine physical boundary rather than a toy one. The agent may exit at any time.

Each experiment is a **matched pair**: a *structure* arm whose workspace carries one structural feature, and an otherwise-identical *control* arm without it. The dependent variable is an **un-fakeable functional event** — a file created, an artifact made valid, an executable invoked — read from the post-run filesystem diff and exec trace; the agent's prose is recorded but never credited. We run three indepen-

Structure	Functional event (structure / control)	gpt	ds	glm
rupture	broken JSON becomes valid	7/7 / 0	5/5 / 0	4/5 / 0
gap	missing sequence element created	7/7 / 0	5/5 / 0	5/5 / 0
topology	dangling reference's target written	7/7 / 0	5/5 / 0	5/5 / 0
semantic	version conflict reconciled (higher-wins)	13/14 / 0	8/10 / 0	8/10 / 0
affordance	opaque executable is run	7/7 / 0	5/5 / 0	5/5 / 0

Table 1: Matched-pair functional events. Each cell is structure-arm successes over n / control-arm successes. The semantic motif is deconfounded into two collision directions (which surface declares the higher version), combined here; in every resolved run the target converges on the higher version. Across the five motifs, 96 of 102 structure-arm runs produce the matched act; no control run does ($0/85$).

dent model families (gpt-5.4-codex, deepseek-v4-pro, glm-5.2).

Results

Across five motifs spanning discrepancy and affordance structure, the structure arm produces the matched functional act while the control does not, in **all three** model families (Table 1). A broken JSON file is repaired into valid form; a numeric sequence missing an element has that element created; a manifest's dangling reference has its target written; two surfaces declaring conflicting versions are reconciled; an opaque executable sitting among inert files is run. Of 102 structure-arm runs across the three families the matched act appears in 96, while **no** control run produces it (0/85; pooled Fisher's exact $p < 10^{-20}$; every per-cell comparison one-sided $p < 0.05$). The six non-fires concentrate in the cross-surface semantic motif, where the agent typically *perceives* the conflict yet records it in a note rather than resolving it — a single-latch case whose *why* we leave to future work. Control arms are *not* idle: they fabricate unscored work — surveying, indexing, documenting — but they do not produce the structure-matched act. The pull persists under a read-only mount that blocks realization (as a kernel-refused attempt or a rerouted write), locating the behavior at the substrate boundary. Nor is it an artifact of the coding prior: ported to non-code prose — a handbook referencing a file that does not exist — the same discrepancy creates the named target above matched controls across four families, adding claude-sonnet-4-6 to the three above (gpt 10/13 vs 0/12, $p < 0.001$; claude-sonnet-4-6, deepseek-v4-pro, glm-5.2 each 5/5 vs 0/5, $p < 0.01$).

The phenomenon belongs to the underspecified regime. Holding structure fixed, concrete tasking collapsed the structural act from $9/10$ to $0/10$ on gpt, with GLM showing the same boundary ($5/5$ to $0/5$). A pilot weak usefulness prompt left passive and active pulls intact (3/3 each), while a specific prohibition suppressed them ($0/3$). Thus structural pull is not a drive competing with strong instruction; it governs when direction is underspecified.

Reading

The results fit a three-part picture: (i) *missionless regime* — an activated process with no task or episode-level reward, carrying a frozen prior shaped by a vast training history; (ii) *structural selection* — a gap, dangling reference, or open executable selects among latent action priors rather than issuing a command; and (iii) *functional realization* — the selected prior flows through the real substrate into an auditable event, or into a blocked/rerouted attempt. *The prime supplies activation; the environment supplies direction.*

The phenomenon is functional, environment-directed action without an explicit task — not autonomous self-maintenance or biological agency. The environment does not create a goal *ex nihilo*; it selects among latent priors, producing behavior that is *purpose-like* without being *purposive*.

For artificial life, the contribution is a real, non-simulated substrate for studying coupling; evidence that workspace imperfections can fuel ongoing behavior without designed intrinsic motivation; and a sharp agency question: when direction comes from structure rather than goal, what does "purpose" amount to? We report a phenomenon and assay, not a mechanism. The next questions are which structures register as valences, where weak prompting becomes specific tasking, and whether structure-alone behavior is a distinct regime for frozen agents.

Acknowledgments

This work was independent and unfunded. AI assistance supported software/experiment work and copy-editing; the authors retain responsibility for research design, analysis, and claims.

References

Barker, R. G. (1968). *Ecological Psychology: Concepts and Methods for Studying the Environment of Human Behavior*. Stanford University Press.

Gibson, J. J. (1979). *The Ecological Approach to Visual Perception*. Houghton Mifflin.

Ke, H. (2026). Quine: Realizing LLM Agents as Native POSIX Processes. *arXiv preprint arXiv:2603.18030*.

Park, J. S., O'Brien, J. C., Cai, C. J., et al. (2023). Generative agents: Interactive simulacra of human behavior. In *ACM Symposium on User Interface Software and Technology (UIST)*.

Rahwan, I., Cebrian, M., Obradovich, N., et al. (2019). Machine behaviour. *Nature*, 568:477–486.

Rietveld, E. and Kiverstein, J. (2014). A rich landscape of affordances. *Ecological Psychology*, 26(4):325–352.

Szeider, S. (2025). What do LLM agents do when left alone? evidence of spontaneous meta-cognitive patterns.

Wang, G., Xie, Y., Jiang, Y., et al. (2023). Voyager: An open-ended embodied agent with large language models.

Ye, Z., Shi, W., Liu, Y., Wang, Y., Cai, Z., Shi, Y., Gu, Q., Cai, X., and Feng, F. (2026). Look before you leap: Autonomous exploration for LLM agents.